Deep Reinforcement Learning

Temporal Difference Learning & Q-learning

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- The TD learning concept
- Prediction using TD learning
- Policy improvement / control using TD learning
  - SARSA
  - Q-learning
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The TD learning concept

Temporal-difference learning and the previous methods

The main aspects of our previous two categories of methods:

1. Monte Carlo (MC): **learning from experience**, possibly without knowledge of a model, e.g., *MC prediction* with exploring starts:

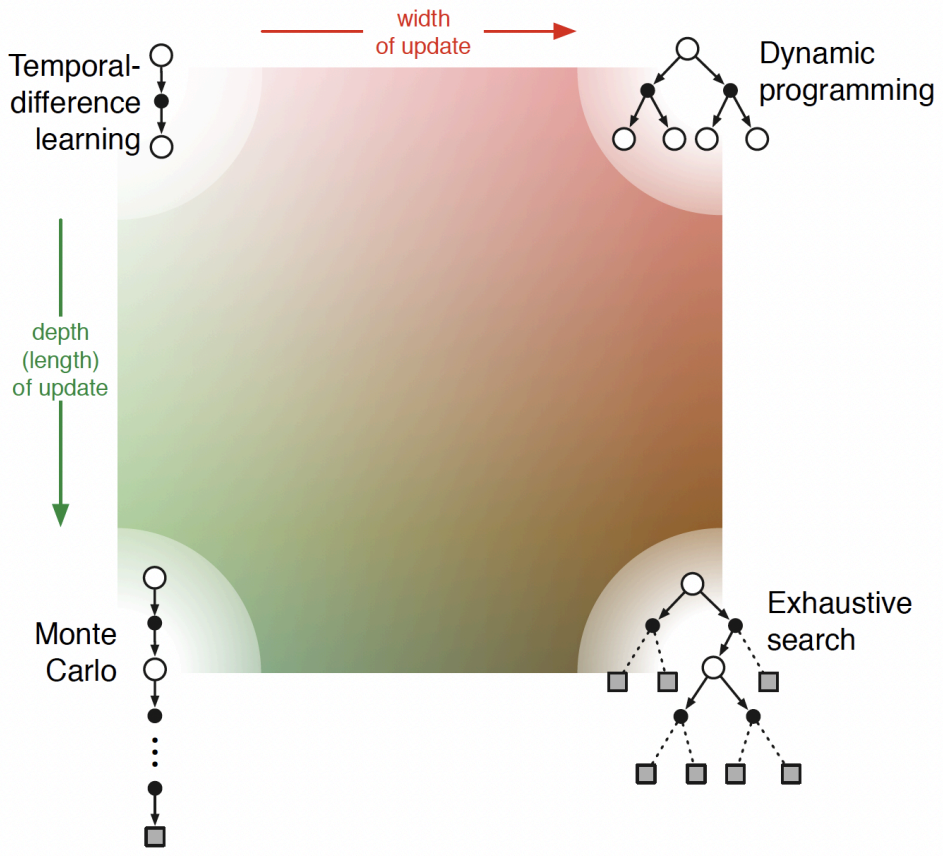
$$Q(s_t, a_t) = Q(s_t, a_t) + \frac{1}{n(s_t, a_t)}\left[g - Q(s_t, a_t)\right].$$

2. Dynamic programming (DP): **bootstrapping** ⇒ updating estimates based on estimates, e.g., *iterative policy evaluation*:

$$V(s) = \sum_{a \in \mathcal{A}} \pi(a|s) \sum_{s' \in \mathcal{S}} p(s'|s, a)\left[r + \gamma V(s')\right].$$

The **concept of temporal difference (TD) learning** is to combine both:

1. *Model-free* prediction and control in unknown MDPs.
2. Updates policy evaluation and improvement in an online fashion (i.e., not per episode) by *bootstrapping*.



(*Sutton and Barto 1998*, *Figure 8.11*)

💡 We are still considering finite MDPs here.

Prediction using TD learning

The general TD prediction update

Recall the **incremental update** of the type we discussed for multi-armed bandits or for MC control $\Rightarrow$ **NewEstimate** $\leftarrow$ **OldEstimate** $+ \mathit{StepSize}$ [**Target** $-$ **OldEstimate**]:

$$V(s_t) \leftarrow V(s_t) + \alpha \left[g_t - V(s_t) \right]. \tag{1}$$

- $\alpha \in (0, 1)$ is the forget factor / step size.
- g_t is the target of the incremental update rule.
- To execute (1), we have to wait until the end of the episode to get g_t.

One-step TD / TD(0) update

$$V(s_t) \leftarrow V(s_t) + \alpha \left[r_t + \gamma V(s_{t+1}) - V(s_t) \right]. \tag{2}$$

- Here, the **TD target** is $r_t + \gamma V(s_{t+1})$.
- TD is *bootstrapping*: estimate $V(s_t)$ based on $V(s_{t+1})$.
- *Delay time of one* time step; no need to wait until the end of the episode.

Algorithmic implementation: TD-based prediction

Algorithm: TD-based prediction.
Parameters: Step size $\alpha \in (0, 1)$

Initialize: $V(s)$ arbitrarily for $s \in \mathcal{S}$, $V(\textbf{terminal}) = 0$

for $k = 1, 2, \ldots, K$ episodes:
    for $t = 0, 1, \ldots, T$:
        Sample action $a_t \sim \pi(a|s)$ and apply
        Observe s_{t+1} and r_t
        $V(s_t) \leftarrow V(s_t) + \alpha\left[r_t + \gamma V(s_{t+1}) - V(s_t)\right]$ (Eq. (2))
        if s_{t+1} is terminal **then** STOP

💡 The algorithm can directly be applied to the prediction of action-value functions.

The TD error and its relation to the MC error

Using the target, we can define the **TD error*** $\delta \Rightarrow$ the expression inside the bracket of Eq. (2) we're using to improve our estimate:

$$\delta_t = \textbf{target} - V(s_t) = r_t + \gamma V(s_{t+1}) - V(s_t).$$

Batch mode: Let's assume that the TD(0) estimate of $V(s)$ does not change within an episode, but that we apply all updates simulatenously once the episode is finished (exactly as we need to do in MC prediction):

$$\begin{aligned}
\underbrace{g_t - V(s_t)}_{\text{MC error}} &= r_t + \gamma g_{t+1} - V(s_t) + \gamma V(s_{t+1}) - \gamma V(s_{t+1}) \\
&= \delta_t + \gamma(\underbrace{g_{t+1} - V(s_{t+1})}_{\text{MC error}}) = \delta_t + \gamma \delta_{t+1} + \gamma^2 (g_{t+2} - V(s_{t+2})) \\
&= \delta_t + \gamma \delta_{t+1} + \gamma^2 \delta_{t+2} + \gamma^3 (g_{t+3} - V(s_{t+3})) = \ldots = \sum_{k=t}^{T} \gamma^{k-t} \delta_k.
\end{aligned}$$

- The MC error is the discounted sum of TD errors in this case.
- If $V(s)$ is updated during an episode (as is common in TD(0)), the above identity only holds approximately.

* For sufficiently small step size α, it can be shown that the TD error conveges to zero in batch mode (Sutton and Barto 1998).

Convergence of TD(0)

Given a finite MDP and a fixed policy π, the state-value estimate V of TD(0) converges to the true V^π (that is, $\lim_{t\to\infty} \delta_t = 0$)

- in the mean for a constant but sufficiently small step-size α and
- with probability one if the step-size satisfies the *Robbins-Munro* condition

$$\sum_{t=1}^{\infty} \alpha_t = \infty \qquad \text{and} \qquad \sum_{t=1}^{\infty} \alpha_t^2 < \infty. \tag{3}$$

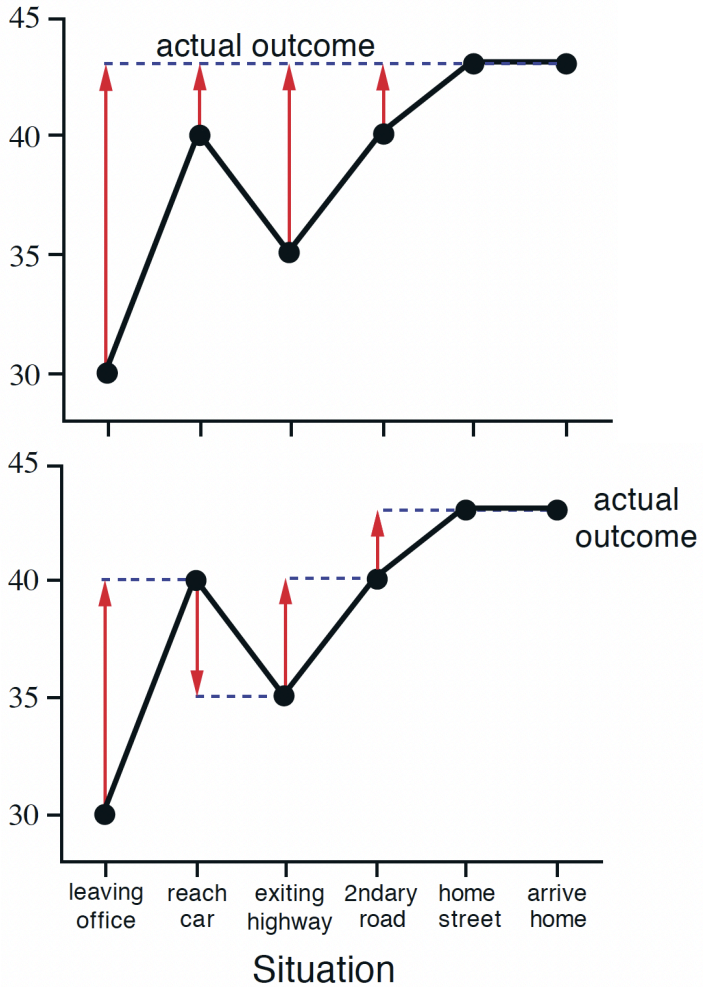
- In particular, $\alpha_k = \frac{1}{k}$ meets the condition (3).
- TD(0) often converges faster than MC, as we will see in the following examples. However, there is no guarantee it's faster.
- TD(0) can be more sensitive to poor initializations $V_0(s)$ compared to MC.

The *driving home* example

Imagine you want to predict your time to get home after work, starting with an estimate of 30 minutes.

State	Elapsed Time (minutes)	Predicted Time to Go	Predicted Total Time
leaving office, friday at 6	0	30	30
reach car, raining	5	35	40
exiting highway	20	15	35
2ndary road, behind truck	30	10	40
entering home street	40	3	43
arrive home	43	0	43

Overview of elapsed time and your predictions of time to go and total time (Sutton and Barto 1998, Example 6.1)



MC vs. TD estimate

- In MC, we estimate the returns g_t **after** we have reached home ⇒ it is hard to assess where the deviation was caused.
- In TD, we can immediately shift our estimate in each time step.

- With TD, we can even learn from continuing tasks!

The batch training AB example (1)

Assume that we have an MDP with only two states: A and B. We do not consider discounting (i.e., $\gamma = 1$).

We collect eight samples from the dynamics, and we would like to estimate the values $V(A)$ and $V(B)$:

Episode k	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
Sequence (s, r, s, r)	A, 0, B, 0	B, 1	B, 1	B, 1	B, 1	B, 1	B, 1	B, 0

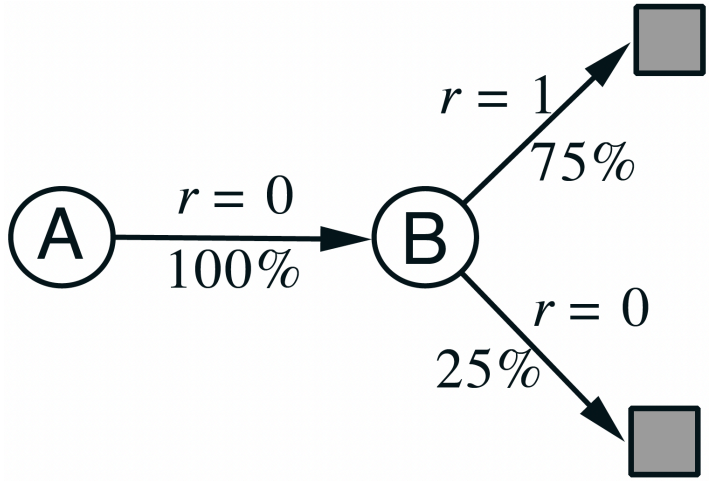
Sample of $K = 8$ trajectories, including the rewards following the states.

Question: What's $V(A)$ and $V(B)$ based on

- the MC estimate?
- the TD(0) estimate?

Before approaching this, let's try to model the underlying MDP!

- B occurs eight times, follwed by $r = +1$ in six, and by $r = 0$ in two cases.
  $\Rightarrow V(B) = \frac{3}{4}$.
- What about $V(A)$?
  1. In our example, the only episode containing A has return $g = 0$ $\quad \Rightarrow V(A) = 0$.
  2. On the other hand, A is followed by $r = 0$ and then $s = B$ $\quad \Rightarrow V(A) = V(B) = \frac{3}{4}$.



Option 2. for modeling the MDP (Sutton and Barto 1998, Ex. 6.4)

Answer 1. is the MC estimate, answer 2. is the TD estimate!

The batch training AB example (2)

Let's formally calculate the MC and TD updates:

$$V(s_t) \leftarrow V(s_t) + \alpha\left[g_t - V(s_t)\right] \qquad \text{(MC)}$$
$$V(s_t) \leftarrow V(s_t) + \alpha\left[r_t + \gamma V(s_{t+1}) - V(s_t)\right] \qquad \text{(TD)}$$

Convergence: no change in the estimate:

$$\alpha\left[g_t - V(s_t)\right] = g_t - V(s_t) = 0 \qquad \text{(MC)}$$
$$\alpha\left[r_t + \gamma V(s_{t+1}) - V(s_t)\right] = r_t + \gamma V(s_{t+1}) - V(s_t) = 0 \qquad \text{(TD)}$$

Consider a batch sweep over the the K episodes:

$$\sum_{k=1}^{K} g_{t,k} - V(s_{t,k}) = 0 \qquad \text{(MC)}$$
$$\sum_{k=1}^{K} r_{t,k} + \gamma V(s_{t+1,k}) - V(s_{t,k}) = 0 \qquad \text{(TD)}$$

The batch training AB example (3)

Apply the previous equations first to **state** B. Since B is a terminal state, $V(s_{t+1}) = 0$ and $g_{t,k} = r_{t,k}$ apply, i.e., the MC and TD updates are identical for B:

$$\text{MC}\Big|_{s=B}: \qquad \sum_{k=1}^{K} g_{t,k} - V(B) = 0 \qquad \Leftrightarrow \qquad V(B) = \frac{1}{K}\sum_{k=1}^{K} r_{t,k} = \frac{6}{8} = \frac{3}{4},$$

$$\text{TD}\Big|_{s=B}: \qquad \sum_{k=1}^{K} r_{t,k} - V(B) = 0 \qquad \Leftrightarrow \qquad V(B) = \frac{1}{K}\sum_{k=1}^{K} r_{t,k} = \frac{6}{8} = \frac{3}{4}.$$

Now consider **state** A, where we have only one trajectory:

- the instantaneous reward following A is zero, and the return of that episode is also zero.
- The TD bootstrap estimate of B is $V(B) = \frac{3}{4}$.

$$\text{MC}\Big|_{s=A}: \qquad \sum_{k=1}^{K} g_{t,k} - V(A) = 0 \qquad \Leftrightarrow \qquad V(A) = 0,$$

$$\text{TD}\Big|_{s=A}: \qquad \sum_{k=1}^{K} r_{t,k} + \gamma V(B) - V(A) = \sum_{k=1}^{K} \gamma V(B) - V(A) = 0 \qquad \Leftrightarrow \qquad V(A) = \gamma V(B) = \frac{3}{4}.$$

Certainty equivalence

Where does this difference come from? $\Rightarrow$ Without going into a detailed derivation:

- MC batch learning converges to the least **squares fit of the sampled returns**:

$$\sum_{k=1}^{K} \sum_{t=1^T} (g_{t,k} - V(s_{t,k}))^2.$$

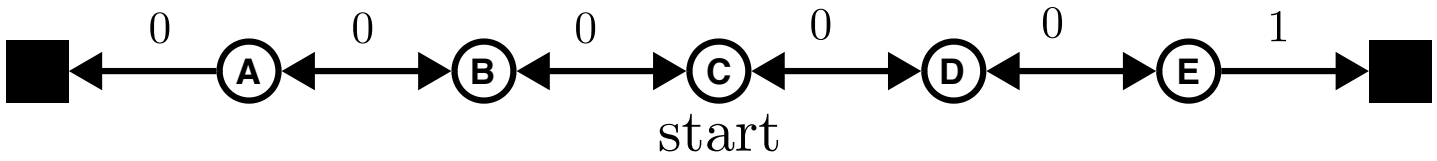
- TD batch learning converges to the **maximum likelihood estimate** such that $(\mathcal{S}, \mathcal{A}, p, r, \gamma)$ explains the data with highest probability:

$$\hat{p}(s'|s,a) = \frac{1}{n(s,a)} \sum_{k=1}^{K} \sum_{t=1^T} \mathbb{1}(s_{t+1,k} = s'|s_{t,k} = s, a_{t,k} = a),$$
$$\hat{r} = \frac{1}{n(s,a)} \sum_{k=1}^{K} \sum_{t=1^T} \mathbb{1}(s_{t,k} = s, a_{t,k} = a) r_{t,k}.$$

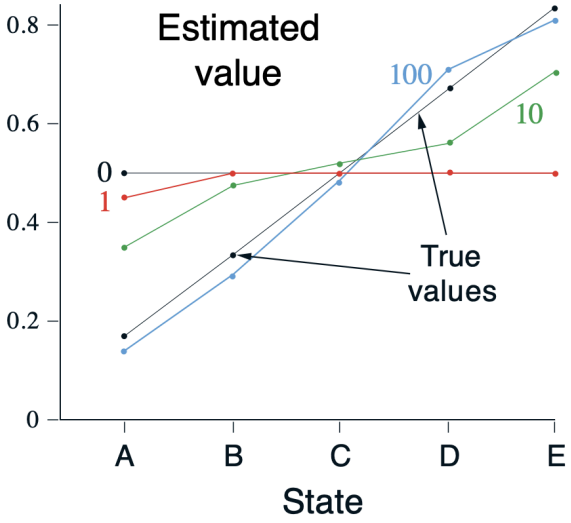
$\Rightarrow$ *TD assumes an MDP problem structure* and is absolutely certain that its internal model concept describes the real world perfectly (so-called **certainty equivalence**).

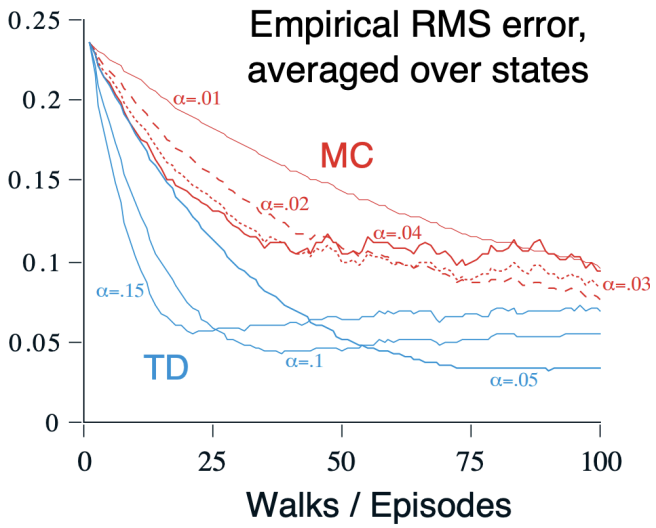
The *random walk* example

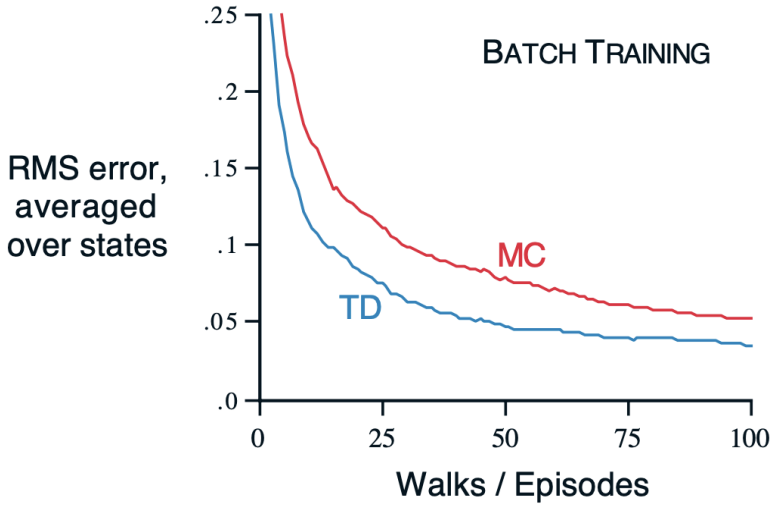
Consider a simple Markov reward process (MRP; no actions), for which we want to estimate $V(s)$ from experience only.



- For all states s, we have $p(\leftarrow) = p(\rightarrow) = 0.5$.
- The MRP is undiscounted ($\gamma = 1$).
- The value of each state is the probability of terminating on the right: $V(A/B/C/D/E) = \frac{1}{6}$ / $\frac{1}{3}$ / $\frac{1}{2}$ / $\frac{2}{3}$ / $\frac{5}{6}$.







TD(0) learning *for the random walk.* ***Left****: Different numbers of iterations (α=0.1).* ***Middle****: Learning curves for different values of α.* ***Right****: Batch training.*

The example is taken from (Sutton and Barto 1998, Ex. 6.2)

Policy improvement / control using TD learning

Application of generalized policy iteration to TD learning

The GPI concept can directly be applied to the TD framework by replacing values with action-values, where we alternate between estimation (i.e., prediction) and improvement:

$$\pi_0 \xrightarrow{E} Q^{\pi_0} \xrightarrow{I} \pi_1 \xrightarrow{E} Q^{\pi_1} \dots \xrightarrow{I} \pi^* \xrightarrow{E} Q^* = Q^{\pi^*}.$$

Estimation: TD(0) update of the Q-function

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \underbrace{\left[\overbrace{r_t + \gamma Q(s_{t+1}, a_{t+1})}^{\text{TD target}} - Q(s_t, a_t)\right]}_{\text{TD error } \delta}. \tag{4}$$

The policy: ϵ-greedy / ϵ-soft*.

Convergence (Sutton and Barto 1998)

- towards Q^* with probability one if all tuples (s, a) are visited infinitely often and the step-size condition (3) is satisfied.
- towards π^* if the policy is GLIE.

Key question: how do we select a_{t+1}? $\Rightarrow$ **On-policy versus off policy!**

* $\pi(a|s) > 0$ for all $s \in \mathcal{S}$ and $a \in \mathcal{A}$.

On-policy TD control: SARSA

s, a, r, s', a': **S**tate-**A**ction-**R**eward-Next **S**tate-Next **A**ction

Algorithm: SARSA.
initialize

- $Q(s, a)$ arbitrarily for $s \in \mathcal{S}, a \in \mathcal{A}$
- $Q(\text{terminal-state}, \cdot) = 0$
- $\pi = \epsilon\text{-greedy}(Q)$

for $k = 1, 2, \ldots, K$ episodes:
  Initialize $s_t \leftarrow s_0, t \leftarrow 0$
  while s_t is not terminal:
    Take action $a_t \sim \pi(s_t)$ and observe (r_t, s_{t+1})
    Select $a_{t+1} \sim \pi(s_{t+1})$
    Update Q given $(s_t, a_t, r_t, s_{t+1}, a_{t+1})$:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left[r_t + \gamma Q(s_{t+1}, a_{t+1}) - Q(s_t, a_t)\right]$$

    Update policy: $\pi = \epsilon\text{-greedy}(Q)$ (This is on-policy!)
    $t \leftarrow t + 1$

Convergence: SARSA for finite-state and finite-action MDPs converges to the optimal action-value,

$$Q(s, a) \to Q^*(s, a),$$

under the following conditions:

1. The policy sequence $\pi_t(a|s)$ is GLIE.
2. The step-sizes α_t satisfy the step-size condition (3) (as does, e.g., $\alpha_t = \frac{1}{t}$).

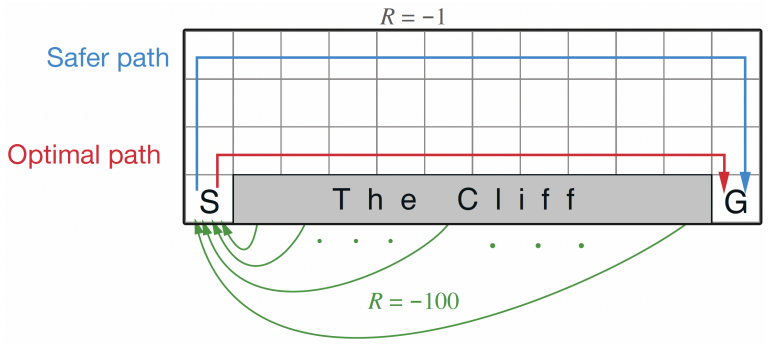
Q-Learning

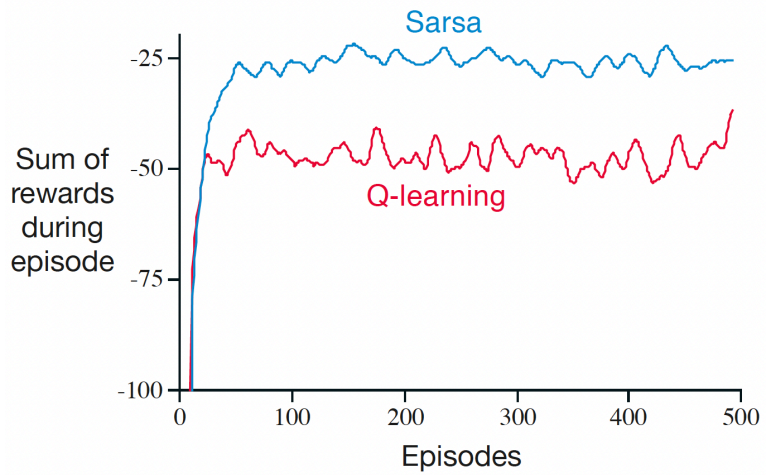
Example: Walking along a cliff

- Usual Gridworld setting.
- All rewards are -1, except
  - the goal G has $r = 0$,
  - the cliff has $r = -100$ and leads back to S.#
- The policy is ϵ-greedy with $\epsilon = 0.1$

Outcome:

- Q-learning learns the optimal path, close along the cliff.
- On-policy SARSA learns a much safer path. *Why do you think that is*?
- In the GLIE case (i.e., $\epsilon \to 0$), both policies will converge to the optimal path.





In our setting, Q-learning is worse than SARSA. *Why do you think that is*?

$\Rightarrow$ the combination of falling off randomly and the resulting very large negative reward.

The example is taken from (Sutton and Barto 1998, Ex. 6.6)

Expected SARSA

What can we do to reduce the large variance in the SARSA update

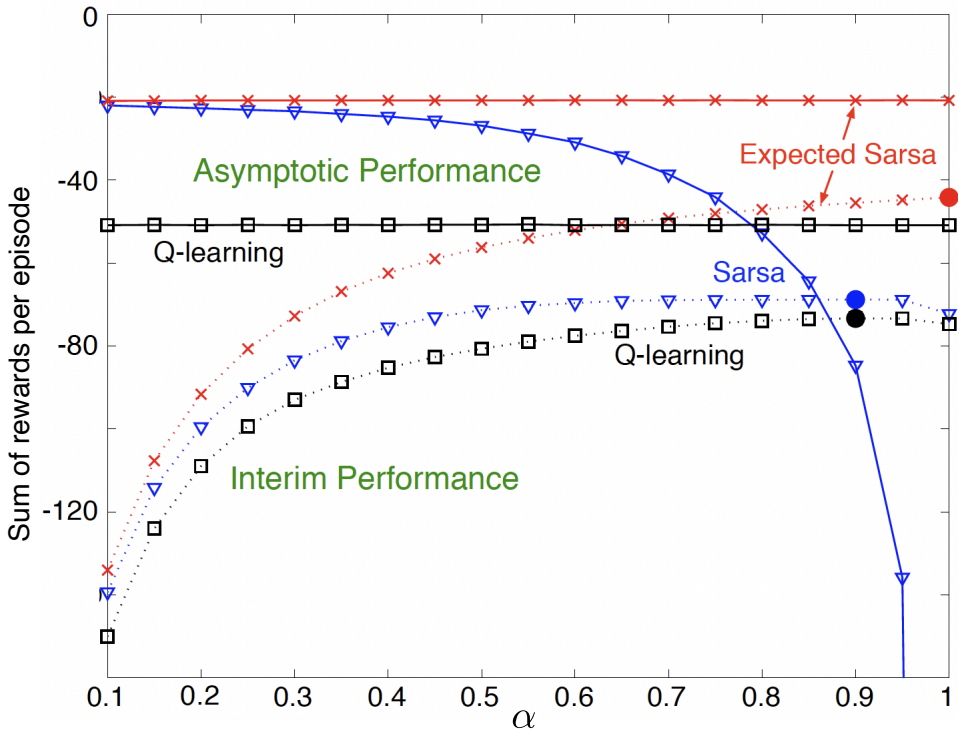
$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha\left[r_t + \gamma Q(s_{t+1}, a_{t+1}) - Q(s_t, a_t)\right]?$$

Let's take the expectation of Q $\rightarrow$ **expected SARSA**:

$$\begin{aligned} Q(s_t, a_t) &\leftarrow Q(s_t, a_t) + \alpha\left[r_t + \gamma \mathbb{E}_\pi\left[Q(s_{t+1}, a) | s_t\right] - Q(s_t, a_t)\right] \\ &= Q(s_t, a_t) + \alpha\left[r_t + \gamma\left(\sum_{a \in \mathcal{A}} \pi(a|s_{t+1}) Q(s_{t+1}, a)\right) - Q(s_t, a_t)\right]. \end{aligned}$$

For comparison: Q-learning

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha\left[r_t + \gamma \max_{\mathbf{a} \in \mathcal{A}} \mathbf{Q}(\mathbf{s_{t+1}}, \mathbf{a}) - Q(s_t, a_t)\right]$$



Double Q-learning

Maximization bias

All control algorithms discussed so far involve *maximization operations*:

- Q-learning: target policy is greedy and directly uses max operator for action-value updates.
- SARSA: typically uses an ϵ-greedy framework $\longrightarrow$ max updates during policy improvement.

This can lead to a significant *positive bias*:

- Maximization over sampled values is used implicitly as an estimate of the maximum value.
- This issue is called maximization bias.

Small example:

- Consider a single state s with multiple possible actions a.
- The true action values are all $Q(s, a) = 0$.
- The sampled estimates $\hat{Q}(s, a)$ are uncertain, i.e., randomly distributed. Some samples are above and below zero.
- Consequence: The maximum of the estimate is positive!

Double Q-learning approach

Split the learning process:

- Divide sampled experience into two sets.
- Use sets to estimate independent estimates $Q_1(s, a)$ and $Q_2(s, a)$.

Assign specific tasks to each estimate:

- Estimate the maximizing action using Q_1:

$$a^* = \arg\max_{a \in \mathcal{A}} Q_1(s, a).$$

- Estimate corresponding action value using Q_2:

$$Q(s, a^*) \approx Q_2(s, a^*) = Q_2(s, \arg\max_{a \in \mathcal{A}} Q_1(s, a)).$$

- Since this approach is perfectly symmetric, we can repeat the process with reversed roles between Q_1 and Q_2.

Double Q-learning algorithm

Algorithm: Q-Learning.
initialize

- $Q_1(s, a)$ and $Q_2(s, a)$ arbitrarily for $s \in \mathcal{S}, a \in \mathcal{A}$
- $Q_1(\text{terminal-state}, \cdot) = Q_2(\text{terminal-state}, \cdot) = 0$

for $k = 1, 2, \ldots, K$ episodes:
  Initialize $s_t \leftarrow s_0$, $t \leftarrow 0$
  while s_t is not terminal:
    Take action a_t using an ϵ-greedy policy on $Q_1 + Q_2$ and observe (r_t, s_{t+1})
    if rand() > 0.5:

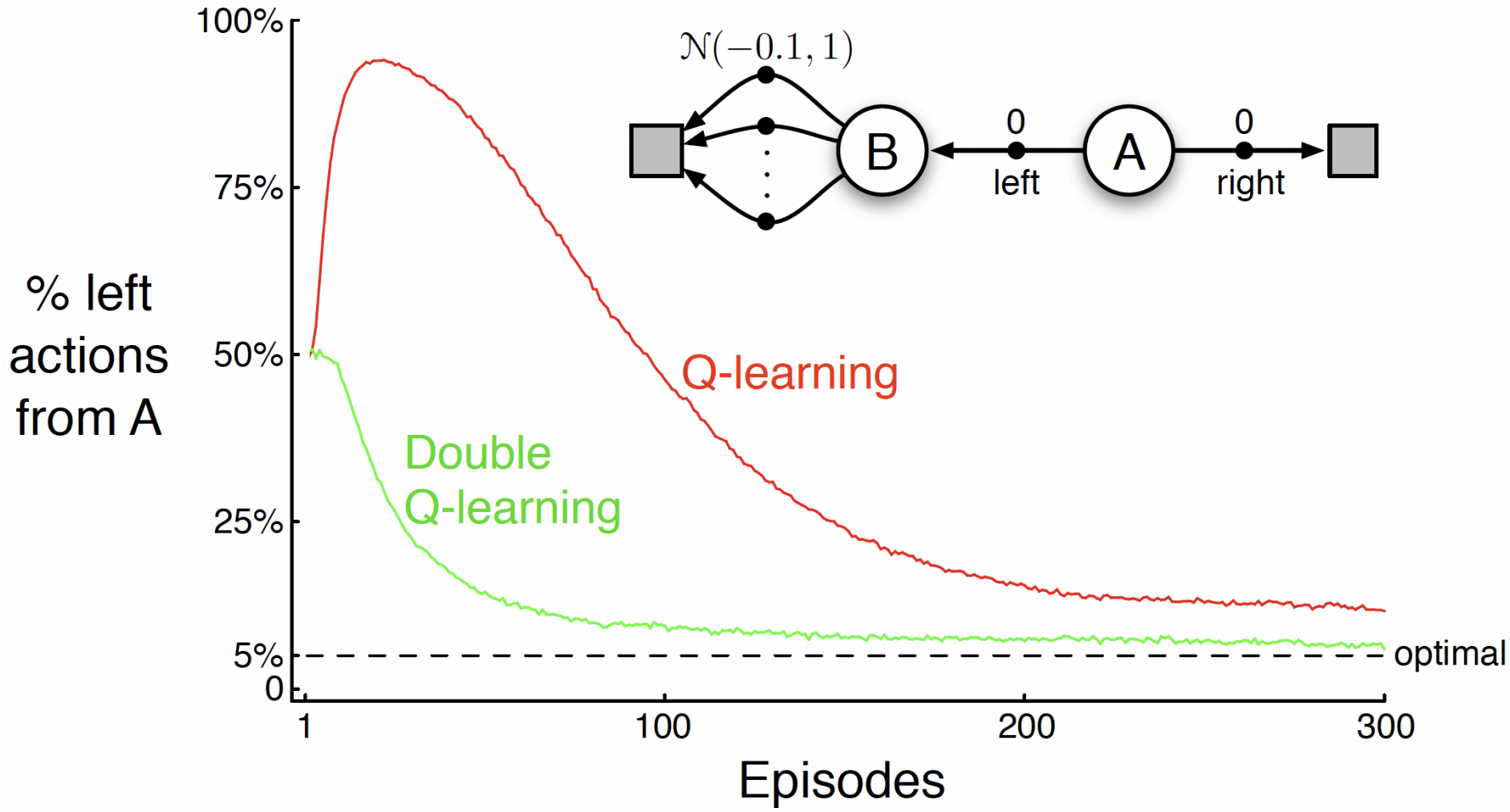
$$Q_1(s_t, a_t) \leftarrow Q_1(s_t, a_t) + \alpha \left[r_t + \gamma Q_2(s_{t+1}, \arg\max_a Q_1(s_{t+1}, a)) - Q_1(s_t, a_t) \right]$$

    else:

$$Q_2(s_t, a_t) \leftarrow Q_2(s_t, a_t) + \alpha \left[r_t + \gamma Q_1(s_{t+1}, \arg\max_a Q_2(s_{t+1}, a)) - Q_2(s_t, a_t) \right]$$

    $t \leftarrow t + 1$

Maximization bias example



Comparison of Q-learning and Double Q-learning on a simple episodic MDP (Sutton and Barto 1998, Ex. 6.7). Q-learning initially learns to take the left action much more often than the right action, and always takes it significantly more often than the 5% minimum probability enforced by ϵ-greedy action selection with $\epsilon = 0.1$. In contrast, Double Q-learning is

essentially unaffected by maximization bias. These data are averaged over 10,000 runs. The initial action-value estimates were zero.

Summary / what you have learned

Summary / what you have learned

- TD unites two key characteristics from DP and MC:
  - From MC: sample-based updates (i.e., operating in unknown MDPs).
  - From DP: update estimates based on other estimates (bootstrapping).
- TD allows certain simplifications and improvements compared to MC:
  - Updates are available after each step and not after each episode.
  - Off-policy learning comes without importance sampling.
  - Exploits MDP formalism by maximum likelihood estimates.
  - Hence, TD prediction and control exhibit a high applicability for many problems.
- Batch training can be used when only limited experience is available, i.e., the available samples are re-processed again and again.
- Greedy policy improvements can lead to maximization biases and, therefore, slow down the learning process.
- TD requires careful tuning of learning parameters:
  - Step size α: how to tune convergence rate vs. uncertainty / accuracy?
  - Exploration vs. exploitation: how to visit all state-action pairs and the optimal policy?

References

Sutton, R. S., and A. G. Barto. 1998. *Reinforcement Learning: An Introduction*. MIT Press.